

Quiz 3 — Testing a Coin: Solutions

M 375T Fall 2026 — 10 points + 2 bonus. Q1: 5 pts; Q2: 5 pts; Bonus: 2 pts.

1. `q.shape` is `(10000,)`: `mean(axis=1)` averages across the 100 columns, leaving one number per row. Each entry $X_{ij} > 0$ is an independent Bernoulli($\frac{1}{2}$) indicator (a standard Gaussian is positive with probability $\frac{1}{2}$), so each row of `q` is the average of 100 fair-coin flips: mean $\frac{1}{2}$ and variance $\frac{1}{4}/100 = 1/400$. Hence `q.mean()` $\rightarrow 0.5$ and `q.std()` $\rightarrow \sqrt{1/400} = 1/(2\sqrt{100}) = 0.05$ as the number of rows grows. *Scoring: 1 pt shape, 2 pts mean, 2 pts standard deviation (1 of the 2 for the variance 0.0025 instead of its square root).* [5 pts]

2. $\varepsilon = 60/100 - 0.5 = 0.1$; $D \approx 2\varepsilon^2 = 0.02$; $T = 2nD = 2 \cdot 100 \cdot 0.02 = 4.0$; $4.0 > 3.84$ is `True`: the test rejects the fair coin at the 5% level. (The exact D is $0.6 \log 1.2 + 0.4 \log 0.8 = 0.0201$, so the approximation is accurate to three decimals.) *Scoring: 1 pt each for ε , D , the Boolean; 2 pts for T . Accept 4 for 4.0. Consistent follow-through after an arithmetic slip keeps the later points.* [5 pts]

Bonus. With exactly 60% heads, $\varepsilon = 0.1$ for every n , so $D \approx 2(0.1)^2 = 0.02$ and $2nD \approx 0.04n$. The test rejects when $0.04n > 3.84$, i.e. $n > 96$. The smallest multiple of five above 96 is $n = 100$. *Scoring: 1 pt for the inequality $0.04n > 3.84$ (or $n > 96$), 1 pt for $n = 100$. Note this concerns an observed 60–40 split, not the sample size that would detect a coin whose true bias is 0.6.* [2 pts]

Q1 mirrors Practice 4 A1 (standardizing a count; the sign of a Gaussian is a fair coin); Q2 mirrors Practice 4 B2–B3 (the quadratic approximation $D \approx 2\varepsilon^2$ and the threshold $2nD > 3.84$); the bonus unwinds the threshold into a rule on n , as B3(c) did for the count k .